Lesson plan · for the teacher

Exploring a dataset

Data science · Level 2 · Lesson 4

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Follow a five-question routine.
- Spot gaps and oddities.
- Write findings with evidence.

Activity: Five questions, three findings

1. Warm-up (5 min): Show a big table of attendance by class and month. Ask: what is the first thing you would look at?
2. Teach (8 min): Walk through the five questions on a small table, drawing a histogram and a scatter plot.
3. Explore (15 min): Each group works through the five questions with its own table, noting oddities.
4. Write (7 min): Groups write three findings with numbers and sources, then share one.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Which question found the most surprises?
- Why check oddities before comparing groups?
- What is your next question?

Exit ticket

- Write one finding from today with a number and a source.

Try it at school tomorrow
Give groups a small school table and the five questions on a card. Each group writes three findings with evidence and one next question.

Practice questions · for you

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1 **Which is the better finding?**

- ◯ Attendance changes during the year
- ◯ Attendance falls to a median of 80% in July
- ◯ Attendance is bad

2 **Put the five exploring questions in order.**

- ☐ Gaps and oddities
- ☐ Groups and relationships
- ☐ What is in it?
- ☐ Centre and spread
- ☐ What shape?

3 **Write summary(xs) so it returns the smallest value, the median and the largest value. Assume xs has an odd number of values.**

```
def summary(xs):
    return 0, 0, 0

print(summary([9, 5, 52, 8, 6]))
```

Write your program:

4 **A column of journey times is skewed right. Which summary should you report?**

- ◯ The median and the range
- ◯ The largest value
- ◯ Only the mean

5 **Which are good findings? Choose all that apply.**

- ☐ Data is important.
- ☐ Median attendance is 80% in July and 95% in March (register, 2025).
- ☐ Our pupils are lazy.
- ☐ Three classes have missing marks for test 2.

✎ **Which school table would you explore first with the five questions? What do you expect to find?**

Check your answers

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1 **Attendance falls to a median of 80% in July**

A good finding has a number, a time and a source.

2 **Top to bottom: 4, 5, 1, 3, 2**

Know the table first, then each column alone, then problems, then comparisons.

3

```
def summary(xs):
    s = sorted(xs)
    return s[0], s[len(s) // 2], s[-1]

print(summary([9, 5, 52, 8, 6]))
```

Sorting puts the values in order, so the first, middle and last items are the minimum, median and maximum.

4 **The median and the range**

Skewed data needs a middle that resists the tail, plus a spread.

5 **Median attendance is 80% in July and 95% in March (register, 2025).; Three classes have missing marks for test 2.**

Good findings are specific and checkable.